

# SMART INDIA HACKATHON 2026

## SOVEREIGN ON-PREMISE AGENTIC AI WORKBENCH

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|--------------------------|--------------------------------------------------------------------------------------------------------------|
| Problem Statement ID:    | SIH26117                                                                                                     |
| Problem Statement Title: | Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work |
| Organization:            | Mangalore Refinery and Petrochemicals Limited (MRPL)                                                         |
| Theme & PS Category:     | Smart Automation   Software                                                                                  |
| Team ID & Name:          | [Registered Team ID] — [Registered Team Name]                                                                |
| Platform & Architecture: | Air-Gapped, Zero-Install Portable Windows/Linux Executable with Embedded Ollama Runtime                      |

**Project Scope Summary:** An industrial-grade, fully sovereign, on-premise desktop AI workbench designed for MRPL refineries. Provides automated multi-turn engineering reasoning, SOP compliance audits, P&ID; drawing inspection, hardware-aware 32-model management, and automatic executive report generation under strict zero-external-network isolation.

# SOVEREIGN AI WORKBENCH — PROBLEM & SOLUTION

## THE PROBLEM

- **Confidential Data Risk:** Sensitive approval notes, P&IDs, Hydrocracker / FCCU telemetry logs, and internal operating code cannot leave refinery premises.
- **Cloud AI Policy Gap:** Commercial cloud AI assistants (OpenAI, Claude) create catastrophic data leakage and cybersecurity compliance violations.
- **Operational Bottleneck:** Manual log verification and cross-referencing against heavy SOP manuals delays maintenance clearance and reduces plant throughput.

## THE PROPOSED SOLUTION

- **100% Self-Hosted & Sovereign:** Fully offline desktop workstation running on refinery hardware with verifiable zero external network traffic.
- **Hardware-Aware Intelligence:** Auto-detects real CPU/RAM/VRAM and scores a 32-model catalog to run the best-fit model comfortably.
- **Multi-Format File Ingestion:** Native extraction from Word, Excel telemetry sheets, PDFs, and P&ID; diagrams.
- **Automated Deliverable Dispatch:** One-click generation of signed executive approval notes in DOCX and PDF directly to Desktop.

## CORE AGENTIC PILLARS & WORKBENCH CAPABILITIES

- 1. Multi-Step Industrial Planning:** AI audits live telemetry against MRPL SOP limits (e.g., MAWP \$150 bar, 6-month PSV calibration window) and formats structured approval notes.
- 2. Local Tool & File Integration:** Universal parser extracts text, tables, and sensor metrics from `.docx`, `.xlsx`, `.pdf`, code, and logs.
- 3. Open-Weight 32-Model Hub:** Categorized repository (Coding, DeepSeek-R1 Reasoning, Vision, General SOP, Enterprise) with 1-click NDJSON streaming downloads and cancellation.
- 4. Scan & Drawing Understanding:** Multimodal visual models (LLaVA, Moondream, Llama-3.2-Vision) inspect P&ID; schematics and equipment gauges.
- 5. VRAM & Process Safety Sandbox:** Windows Job Object (`KILL_ON_JOB_CLOSE`), orphan cleanup, and `keep_alive=0` memory release.
- 6. Executive Deliverable Export:** Styled ReportLab PDF and python-docx approval note generator dispatched directly to OS Desktop.

# TECHNICAL APPROACH & ARCHITECTURE

## 1. LAYERED LOCAL WORKBENCH STACK

- **Presentation Layer:** Single-Page Desktop App in PyWebView (HTML5/CSS3/JS); 5 luxury preset themes (Midnight Obsidian, Titanium, Azure, Matrix, Amethyst) + custom palette builder; real-time stop generation button.
- **Backend Server (main.py):** Multithreaded TCP Server on port 8085; REST API + NDJSON streaming progress endpoints; daemon thread dispatch.
- **Core Engine (sovereign\_engine.py):** Hardware detection, 32-model catalog scorer, multi-turn chat orchestrator, universal file parser.
- **Process & Memory Supervisor:** Win32 Job Object (``0x2000``), startup orphan process killer, dynamic VRAM model unloader (``keep_alive=0``).
- **Inference Layer:** Embedded portable Ollama daemon (``port 11434``) running quantized GGUF weights on CUDA GPU / CPU AVX2.
- **Air-Gap Isolation:** Hardcoded loopback (``127.0.0.1``), zero outbound calls, standalone ``data/`` directory structure.

## 2. STREAMLINED INSPECTION & APPROVAL WORKFLOW

- Step 1: Telemetry & Report Ingestion:** Operator attaches Hydrocracker / Crude Distillation logs, Excel sensor data, or P&ID; drawings.
- Step 2: Context Parsing:** Universal parser extracts operating pressures (142.5 bar), temperatures (410°C), and valve tags (PSV-102A).
- Step 3: Hardware-Guided Model Routing:** Engine matches prompt complexity to the optimal installed model (e.g., DeepSeek-R1 / Qwen2.5).
- Step 4: SOP Compliance Audit:** LLM checks operating pressure against MRPL SOP limits (MAWP: 150 bar, 140 bar threshold audit).
- Step 5: Executive Deliverable Generation:** Auto-formats structured findings with executive summary, compliance status, and signature lines.
- Step 6: Direct Desktop Dispatch:** Builds formatted ``docx`` and styled ``pdf`` documents directly on Desktop and launches in Word.

**Implementation Components:** **Agent Harness:** Multithreaded Python (``main.py`` + ``sovereign_engine.py``) | **Inference:** Embedded Portable Ollama + GGUF Quantized Models (CUDA / CPU AVX2) | **File Ingestion:** Native ``python-docx``, ``openpyxl``, PDF & text parsers | **UI Framework:** PyWebView + Luxury Responsive HTML5 SPA (5 Themes) | **Safety & Isolation:** Win32 Job Object (``KILL_ON_JOB_CLOSE``) + Dynamic VRAM Unloader (``keep_alive: 0``) + Zero-Egress Loopback

FEASIBILITY, VIABILITY & RISK MITIGATION

|                                                                |                                                          |                                                                   |                                                               |                                                            |                                                           |
|----------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|
| <b>1. Foundation</b><br>Portable EXE, server & embedded engine | <b>2. Model Hub</b><br>32 models, 1-click streaming pull | <b>3. Guardrails</b><br>Job Object, VRAM unloader & orphan killer | <b>4. Agentic Core</b><br>Multi-turn memory & MRPL SOP prompt | <b>5. File Ingestion</b><br>DOCX, XLSX, PDF & P&ID; parser | <b>6. Deliverables</b><br>ReportLab PDF & Word to Desktop |
|----------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------|

| Single Workstation Feasibility | Implementation & Verification                                                                                   | Identified Risk              | Control Implemented                                                                    | Live Demo Evidence                                                              |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Open-Weight Models             | Curated 32-model catalog (0.135B to 70B) covering Qwen2.5, DeepSeek-R1, Llama-3.2, Gemma-2, Moondream, Mistral. | Model Exceeds VRAM (OOM)     | Hardware profiler + formula: $\text{Mem} = \text{Params} \times 0.65 + 0.8 \text{ GB}$ | Dynamic compatibility badges (■ Optimal, ■ Full GPU, ■ CPU) prevent OOM.        |
| Quantized GGUF                 | 4-bit/8-bit quantization fits low-end GPUs (2GB VRAM GT 710) up to multi-GPU high-memory servers.               | VRAM Leak / Zombie Processes | Win32 Job Object (0x2000) + dynamic keep_alive: 0 model unloader                       | Verified via nvidia-smi: exactly 0 MB VRAM retained after exit or model switch. |
| Embedded Runtime               | Standalone Ollama daemon with isolated storage (data/models/); zero admin rights or installs needed.            | Data Leakage via Cloud       | Hardcoded loopback (127.0.0.1) with zero external calls                                | Wireshark network monitor records 0 outbound packets during chat and export.    |
| Portable Storage               | All settings, sessions, models, and exports stored in portable data/ folder packaged in single ~21MB ZIP.       | Interrupted Downloads        | NDJSON streaming reader with thread cancellation event                                 | Live byte progress bar with instant responsive <b>Cancel</b> button.            |

# IMPACT, BENEFITS & INDUSTRIAL VALUE

## WORKFLOW TRANSFORMATION

- ■ **Gear 1: Profile & Route:** Hardware profiler auto-detects GPU/RAM and routes task to optimal model from 32-model hub.
- ■ **Gear 2: Read & Ingest:** Universal parser extracts telemetry metrics from Excel sensor logs, DOCX, and P&ID; drawings.
- ■ **Gear 3: Audit in Sandbox:** Local LLM evaluates operational parameters against MRPL SOP limits (MAWP threshold, valve dates).
- ■ **Gear 4: Deliver & Clear:** Exports signed executive approval note to Desktop and releases VRAM immediately.

## WHY IT MATTERS

- **Absolute Confidentiality:** Telemetry logs, tariff accounting, and internal memos never leave refinery boundaries.
- **Zero-Friction Portability:** Single ~21MB ZIP runs out-of-the-box on Windows/Linux without admin privileges or pip installs.
- **Hardware Inclusivity:** Runs on modest 2GB VRAM cards (GT 710) up to multi-GPU enterprise AI workstations.

## STAKEHOLDER VALUE MATRIX (BEFORE VS. AFTER)

### ■ ■ ■ Plant & Operations Engineers:

- Before:* Manual spreadsheet parsing & cross-referencing against 200+ page SOP manuals.
- After:* Instant file ingestion, automated telemetry audits, and evidence-backed draft notes.

### ■ ■ Operations & Safety Approvers:

- Before:* Slow, multi-day document review cycles causing operational delays.
- After:* Structured, standardized, and traceable compliance memos generated in seconds.

### ■ IT & Cybersecurity Officers:

- Before:* Massive risk of proprietary refinery data leakage via public cloud LLMs.
- After:* 100% on-premise air-gapped execution with verifiable zero egress traffic.

### ■ Executive Management:

- Before:* High recurring cloud API costs, vendor lock-in, and compliance penalties.
- After:* Zero license costs, offline USB-deployable solution across all refinery units.

# RESEARCH, REFERENCES & IMPLEMENTATION ROADMAP

## LIVE DEMONSTRATION CHECKLIST

- [x] **Hardware Auto-Profiling:** Accurately detects CPU cores, RAM, free disk, and GPU VRAM via nvidia-smi and auto-recommends optimal model.
- [x] **Interactive 32-Model Hub:** 1-click NDJSON streaming download with real-time progress bar and responsive cancel support.
- [x] **Multi-Format Context Ingestion:** Extracts and injects text from Word documents, Excel telemetry sheets, PDFs, and code files into the prompt.
- [x] **Executive Deliverable Dispatch:** One-click generation of styled ReportLab PDFs and formatted Microsoft Word (`.docx`) approval memos directly to Desktop.
- [x] **100% VRAM & Process Safety:** Windows Job Object ('KILL\_ON\_JOB\_CLOSE') + `keep\_alive: 0` model unloading ensures 0 MB VRAM leakage and zero orphan processes on exit.

## 10-STAGE IMPLEMENTATION ROADMAP

1. **SIH Scope:** Defined air-gapped, multimodal sovereign AI scope for MRPL refinery.
2. **Portable Architecture:** Built self-contained data/ folder structure and standalone EXE.
3. **Embedded Inference:** Bundled standalone Ollama daemon with isolated OLLAMA\_MODELS.
4. **Universal Ingestion:** Native python-docx, openpyxl, PDF and diagram text extractors.
5. **Strict Air-Gap Proof:** Hardcoded loopback (127.0.0.1) with zero outbound calls.
6. **Hardware Auto-Scoring:** Memory formula scoring with dynamic visual fit badges.
7. **Curated 32-Model Hub:** Categorized catalog with 1-click download and live progress bars.
8. **VRAM & Process Guard:** Win32 Job Object supervisor and dynamic keep\_alive: 0 unloader.
9. **Document Generator:** ReportLab PDF formatting and direct desktop Word dispatch.
10. **Luxury Modern UI:** 5 preset themes, custom palette builder, and stop response controller.

**Technical References & URLs:** **SIH Portal:** [sih.gov.in/sih2026PS](https://sih.gov.in/sih2026PS) (Problem Statement SIH26117) | **Inference Engine:** [ollama.com](https://ollama.com), [github.com/ollama/ollama](https://github.com/ollama/ollama), [github.com/ggml-org/llama.cpp](https://github.com/ggml-org/llama.cpp) | **Parsing & Docs:** [github.com/python-openxml/python-docx](https://github.com/python-openxml/python-docx), [github.com/openpyxl/openpyxl](https://github.com/openpyxl/openpyxl), [github.com/reportlab/reportlab](https://github.com/reportlab/reportlab) | **Open-Weight Models:** [github.com/qwenlm/Qwen2.5](https://github.com/qwenlm/Qwen2.5), [github.com/deepseek-ai/DeepSeek-R1](https://github.com/deepseek-ai/DeepSeek-R1), [github.com/meta-llama/llama-models](https://github.com/meta-llama/llama-models), [github.com/vikhyat/moondream](https://github.com/vikhyat/moondream)